Suleyman Yildirim

MSc | AI Research Engineer | Deep Learning, Computer Vision, Robotics

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Profile

AI Research Engineer with 4 years of experience building, training, and evaluating deep learning systems across computer vision, generative AI, robotics, time-series forecasting, and autonomous systems. Experienced in Python/PyTorch implementation, multi-GPU training on Linux/SLURM environments, large-scale ML experimentation, data preprocessing, feature engineering, and deployment-oriented testing. Combines academic research with industrial R&D experience at Huawei, Baykar, and Koç University/KUIS AI, with a strong focus on translating research ideas into reliable, high-performance AI systems.

Knowledge & Skills

AI Research & Deep Learning — Machine Learning, Deep Learning, Computer Vision, Generative Models, Transformers, Reinforcement Learning, Image Processing, Time-Series Forecasting, LLMs

Data & Experimentation — Data Preprocessing, Feature Engineering, Exploratory Data Analysis, Statistical Analysis, Model Training, Evaluation, Experimentation

Programming — Python, PyTorch, NumPy, Pandas, SciPy, MATLAB, C++, Git, Linux, SLURM, Multi-GPU

Systems, Robotics & Simulation — ROS, PX4, MAVLink, Gazebo, SITL Testing, Real-World Drone Testing

Education

MSc - Computer Sciences and Engineering, 09/2022 – 12/2025
Koç University (100% English)
- KUIS AI Fellow

BSc - Electrical and Electronics Engineering, 09/2017 – 06/2022
İstanbul Medipol University (100% English)
- Full Scholarship, High Honor Student
- Graduated **3rd in faculty** with **GPA: 3.81/4.00**.
- Undergraduate thesis funded by **TÜBİTAK** .

Experience

Robotics AI Engineer, *DroneLeaf AI Solutions* 09/2024 – Present | Abu Dhabi, UAE
- Implemented core flight-controller modules in C++, contributing to reliable autonomous drone control in real-world environments.
- Developed a modular Python SDK for high-level mission planning, enabling structured definition, execution, and management of complex drone missions.
- Built simulation-to-real validation workflows using Gazebo SITL to test system behaviour before physical deployment.
- Conducted end-to-end real-world testing on drones of varying sizes in indoor OptiTrack and outdoor GPS environments, validating robustness, reliability, and deployment readiness.

Technologies: ROS, PX4, MAVLink, C++, Python, Gazebo

AI Research Engineer, *Huawei* 04/2023 – 09/2024 | Istanbul, Turkiye

Turkey R&D Center, Wireless Department

- Developed machine learning and deep learning models for long-term and short-term cellular network traffic forecasting using large-scale 5G network datasets.
- Performed data preprocessing, exploratory data analysis, statistical analysis, and feature engineering to support robust ML experimentation.

Technologies: Python, PyTorch, Pandas, NumPy, SciPy, Darts

AI Research Fellow, *Koç University & İş Bank Artificial Intelligence Research Center (KUIS AI)* 09/2022 – 12/2025 | Istanbul, Turkiye

Multimedia, Vision and Graphics Laboratory (MVGL)

- Conducted research on blind image super-resolution using real-world image data, focusing on robust deep learning models for challenging computer vision problems.
- Leveraged generative models including GANs, diffusion models, and MAGE to enhance image quality and generate realistic low-resolution image datasets.
- Implemented, trained, and evaluated PyTorch-based deep learning models on GPU/Linux/SLURM environments.
- Collaborated with academic researchers to develop, test, and improve model architectures and experimental pipelines.

Technologies: Python, PyTorch, SLURM, Linux, Multi-GPU

AI Research Engineer, *Baykar Technology* 10/2021 – 07/2022 | Istanbul, Turkiye

Smart Systems and High Level Autonomy Team

- Researched and implemented state-of-the-art deep reinforcement learning algorithms for autonomous flight control in simulation environments.
- Developed a modular Python library for experimentation with DRL models, improving research iteration, reproducibility, and testing efficiency.
- Engineered a real-time C++ path-planning algorithm for MALE-class UAVs, reducing computation time for large-scale mission scenarios.

Technologies: Python, PyTorch, C++, OpenAI Gym, Linux

Publications

- **Yildirim, S.,** Erdem, A., Erdem, E**.** *“Generative Augmentation for Real-World Image Super-Resolution.”* Manuscript in preparation for submission to *ICLR 2027.*
- **Yildirim, S.**, Ates, H.F., Gunturk, B.K. *“Deep Learning-Based Blind Image Super-Resolution with Iterative Kernel Reconstruction and Noise Estimation.” Computer Vision and Image Understanding (CVIU)*, 2023.
- **Yildirim, S.**, Ates, H.F., Gunturk, B.K. *“Iterative Kernel Reconstruction for Deep Learning-Based Blind Image Super-Resolution.” IEEE International Conference on Image Processing (ICIP)*, 2022.
- Özbek, M., **Yildirim, S.**, Aksoy, M., Kernin, E., Koyuncu, E. *“Harfang3D Dog-Fight Sandbox: A Reinforcement Learning Research Platform for the Customized Control Tasks of Fighter Aircrafts.” arXiv e-prints*, 2022.

Awards and Honors

- Ranked **3rd in Faculty**, High Honor Student – Istanbul Medipol University, 2022
- **TÜBİTAK Research Grant** for Undergraduate Thesis, 2021–2022
- **Top 8 Worldwide** – CanSat Competition by American Astronautical Society, 2020
- **2nd Place** – Siemens Healthineers IDMedTech Innovation Competition, Turkey, 2020

Languages

English, *C1*

- **IELTS** 7.5 / **Duolingo** 140